

# Time-Step-Free Physics-Based Modeling of Retention Loss in Cryogenic 3-D NAND Flash

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**Abstract**—We propose a time-step-free physics-based model for retention loss in cryogenic 3-D NAND flash memory. By solving the first-order autonomous detrapping equation in closed form, the proposed model directly evaluates  $\Delta V_{th}(t)$  from the initially programmed trapped charge distribution without transient time stepping. For a fixed number of targeted retention times, the computational cost is independent of the final physical retention time, enabling efficient short- and long-term retention prediction. The model is benchmarked against TCAD simulations calibrated to the time scale of reported cryogenic retention data and configured to isolate detrapping loss by suppressing thermally activated transitions during retention. It reproduces the TCAD predicted retention transient and tunneling oxide thickness dependence for 3-5 nm while reducing the average runtime from  $>25000$  s to  $<4$  s.

**Index Terms**—time-step-free modeling, autonomous ODE, detrapping, 3-D NAND flash memory, retention loss modeling

## I. INTRODUCTION

THE rapid increase in data storage demand, driven by emerging computing technologies such as artificial intelligence (AI) and high-performance computing (HPC), has raised significant concerns about the reliability of 3-D NAND flash memories, particularly regarding retention loss [1].

As storage capacity per cell increases, ensuring long-term stability of each bit's threshold voltage distribution over time scales from days to years becomes increasingly critical. Moreover, prior works have shown that trapped charge detrapping can still occur at cryogenic temperatures [2], [3], supporting models that treat detrapping as temperature independent [4].

In this study, a time-step-free physics-based model is developed for retention loss based on tunneling emission mechanisms in 3-D NAND flash memory. The proposed simulation framework uses the initially programmed trapped charge distribution as input and enables highly efficient computation by using closed-form solution of the autonomous detrapping equation, eliminating sequential transient time stepping.

## II. MODELING

This section presents the modeling framework for time-step-free simulation of retention loss. First, a cylindrical Poisson solution is derived for the oxide–nitride–oxide (O/N/O)

gate stack, incorporating quantized multiple layers within the charge trap nitride (CTN) layer. After obtaining the detailed Poisson solution for these discretized layers, the threshold voltage shift ( $\Delta V_{th}$ ) is calculated using a determinant based method. Finally, based on previous studies of detrapping-induced emission rates [4], an autonomous equation governing retention loss behavior is proposed.

### A. Detailed Poisson Solution

An electrical potential distribution obtained from the cylindrical Poisson equation for discretized layers in the ONO structure can be expressed as follows [4], [5].

$$\begin{cases} V_{Si}(r_{Si}) = V_{ch} \\ V_{TOX}(r) = V_{ch} + \frac{K_{TOX}}{\epsilon_{TOX}} \ln \frac{r}{r_{Si}} \\ V_{CTN,i}(r) = V_{CTN,i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{CTN}} \ln \frac{r}{r_{i-1}} \\ V_{BOX}(r) = V_{CTN,N}(r_N) + \frac{K_{BOX}}{\epsilon_{BOX}} \ln \frac{r}{r_N} \\ V_{BOX}(r_{BOX}) = V_G \end{cases} \quad (1)$$

where  $i = 1-N$  denotes the index of each discretized slice of the CTN layer in the radial direction, and  $N$  is the total number of radial partitions.

Solving  $\mathbf{x}$  vector can calculate the total  $N + 2$  potential parameters.

$$\mathbf{x} = [K_{TOX} \quad K_1 \quad \cdots \quad K_N \quad K_{BOX}]^T \quad (2)$$

$$A\mathbf{x} = \mathbf{b} \quad (3)$$

Matrix  $A$  and vector  $\mathbf{b}$  are constructed by enforcing a total of  $N+1$  transverse electric field continuity conditions ( $\epsilon_1 \mathcal{E}_1 = \epsilon_2 \mathcal{E}_2$ ) together with an additional boundary condition defined by reference potential.

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{TOX}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{BOX}} \end{bmatrix} \quad (4)$$

$$\mathbf{b} = [\frac{\Delta Q_0}{2\pi} \quad \frac{\Delta Q_1}{2\pi} \quad \cdots \quad \frac{\Delta Q_N}{2\pi} \quad \Delta V]^T \quad (5)$$

$$\Delta Q_i = \pi r_i^2 (n_{t,i} - n_{t,i-1}) \quad (6)$$

$$\Delta V = (V_G - V_{ch}) - \sum_{i=1}^N \frac{qn_{t,i}}{4\epsilon_{CTN}} (r_i^2 - r_{i-1}^2) \quad (7)$$

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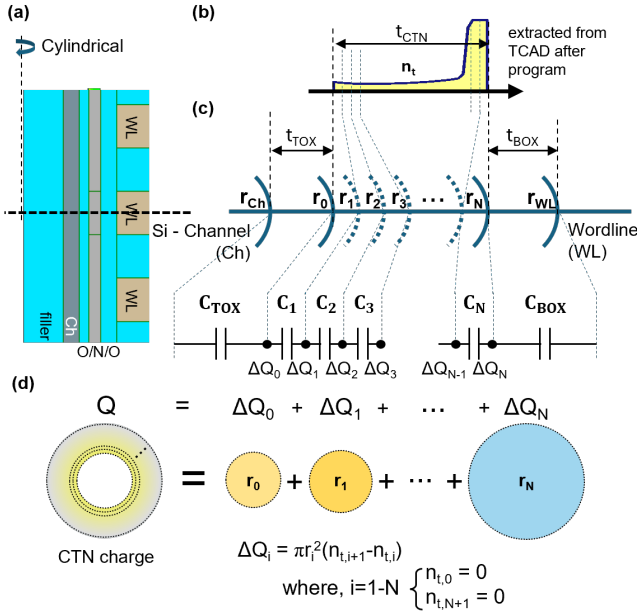


Fig. 1. (a) shows a radial slice of 3-D NAND flash memory, (b) shows the programmed TCAD-extracted post-program trapped charge distribution, (c) shows capacitance and node charge representation, and (d) shows the visualization of  $\Delta Q_i$  for each node.

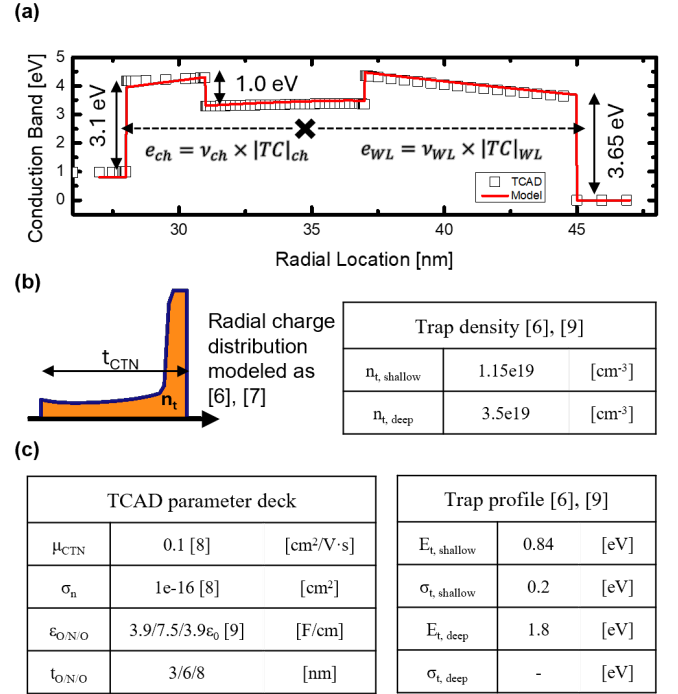


Fig. 2. (a) The band diagram with barrier notations where detrapping of trapped electron occurs, (b) a detail programmed trapped charge distribution after program operation the radial trapped charge distribution used as the model initial condition, and (c) TCAD parameter deck and used trap profile within the simulation [6]–[9].

$C_{\text{TOX}}$ ,  $C_i$ , and  $C_{\text{BOX}}$  in (4) denote the capacitances of each cylindrical layer, as illustrated in Fig. 1(c).  $\Delta Q_i$  in (5) represents the charge component associated with the cylindrical shell at radius  $r_i$ , which can be superpositioned as  $Q = \sum \Delta Q_i$  as demonstrated on Fig. 1(d). Furthermore, (5) indicates that  $b(\Delta Q_i, V_G - V_{\text{ch}})$  varies with the trapped charge distribution and the applied voltage conditions.

### B. Q - $\Delta V_{\text{th}}$ Relation

For the  $\Delta V_{\text{th}}$  calculation, the gate voltage difference required to maintain equal channel field conditions must be determined [4], [5].

$$F_{\text{TOX}}(r_{\text{Si}}) = \frac{K_{\text{TOX}}}{\epsilon_{\text{TOX}}} \frac{1}{r_{\text{Si}}} \quad (8)$$

To achieve equal channel field conditions, the same  $K_{\text{TOX}}$  value under the given conditions must be established. By Cramer's rule, the  $K_{\text{TOX}}$  parameter can be solved through determinant calculation.

$$K_{\text{TOX}} = \frac{\det [A_{\text{b}(\Delta Q_i, \Delta V_{\text{th}})}]}{\det A} \quad (9)$$

where notation  $A_{\text{b}(\Delta Q_i, \Delta V_{\text{th}})}$  stands for 1<sup>st</sup> column of matrix  $A$  replaced with vector  $\text{b}(\Delta Q_i, \Delta V_{\text{th}})$ .

Therefore,  $\Delta V_{\text{th}}$  can be solved easily by using the property of multilinearity.

$$\det [A_{\text{b}(0, \Delta V_{\text{th}})}] = -\Delta V_{\text{th}} = \det [A_{\text{b}(\Delta Q_i, 0)}] \quad (10)$$

$$\therefore \Delta V_{\text{th}} = -\det [A_{\text{b}(\Delta Q_i, 0)}] \quad (11)$$

The threshold voltage difference can be calculated with single determinant computation with initially programmed trapped charge distribution is given as Fig. 2(b).

### C. Time-step-free Detrapping Modeling

With the given initial conditions, an autonomous equation is established for each cell, divided by radial position ( $r$ ) and trap depth ( $E_t$ ).

$$\frac{d}{dt} n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}}) n_t(r, E_t, t) \quad (12)$$

Then, detrapping relation can be express as following autonomous ODE solution.

$$n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \quad (13)$$

The emission rate for trapped electron can be computed as following equations based on a previous study [4].

$$e = \nu \times |TC| \quad (14)$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m_0 \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}} \quad (15)$$

$$|TC| = \prod \exp \left( -\frac{4\sqrt{2m^*} \Phi_H^{1.5} - \Phi_L^{1.5}}{3\hbar q F} \right) \quad (16)$$

for both channel and wordline directions as demonstrated in Fig. 2(a).

This autonomous detrapping equation (13) is the key relation that enables a simulation to directly compute the result at a specific time once initially programmed trapped charge distribution is specified. Therefore, for a fixed spatial/energy discretization and a fixed number of evaluated retention times, the computational cost is governed by the number of trap bins rather than by the final physical retention time.

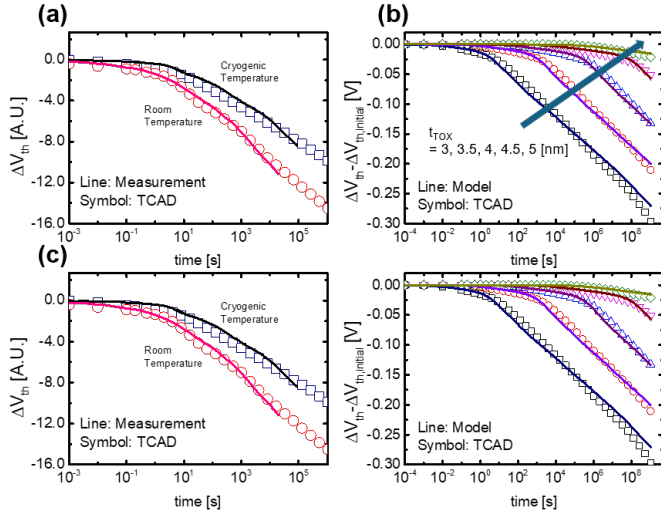


Fig. 3. (a) The TCAD simulation calibration with the measurement from previous study [2]. (b) A detail detrapping curve variation over time with  $t_{TOX}$  from 3 nm to 5 nm.

TABLE I

COMPARISON BETWEEN A TCAD AND TIME-STEP-FREE MODEL

| Property                               | TCAD                                            | Time-step-free model                       |
|----------------------------------------|-------------------------------------------------|--------------------------------------------|
| Temporal treatment                     | sequential time stepping                        | closed-form evaluation of eq (13)          |
| Dependence on final physical time      | Runtime increases with the final retention time | Runtime independent of final physical time |
| Run Time (s)<br>(average of Fig. 3(b)) | >25000                                          | <4                                         |

### III. SIMULATION AND RESULT

According to [2], cryogenic conditions enable the detrapping component to be isolated from retention loss separated from thermally activated retention-loss components. However, direct TCAD simulation at cryogenic temperature about 4.2 K condition did not converge reliably in our setup. Therefore, before applying the cryogenic-emulation condition, two convergence tendencies were compared in the extreme condition: one obtained by lowering the temperature toward the lowest convergent cryogenic range, and the other obtained by reducing the electron capture cross section toward zero. Both approaches showed the same retention-time tendency in the extreme condition, indicating that the suppression of thermally activated transitions can reproduce the dominant cryogenic detrapping behavior within the convergent TCAD framework. Therefore, to emulate cryogenic conditions during retention operation within the TCAD framework, the capture cross section was adjusted to an extremely low value ( $\sigma_n \rightarrow 0$ ) only during retention. This manipulation suppresses thermal electron transitions between trapped and conduction states within the CTN layer, while maintaining physical consistency in other mechanisms [10]. Based on this convergence tendency, the calibration in Fig. 3(a) was performed using the original time-scale matching strategy, without introducing an additional calibration figure. Thus, Fig. 3(a) should be interpreted as a calibrated cryogenic-emulation benchmark rather than a direct

#### 4.2 K TCAD simulation.

The TCAD benchmark was built using Synopsys Sentaurus TCAD following the Device manual [10]. The retention deck includes thermionic emission, Poole-Frenkel emission, and nonlocal tunneling, providing a TCAD reference with thermal and tunneling detrapping mechanisms. The TCAD deck shown in Fig. 2(c) was calibrated by matching the retention time scale of the measurement in previous study [2]. Since  $\Delta V_{th}$  is reported in arbitrary units, the calibration was done by matching time scale only. Program operation was performed by single 1  $\mu$ s pulse and calibrated to obtain  $\Delta V_{th} = 1.8$  V. To properly focus on the detrapping effect from the CTN layer, bandgap-engineered processes were neglected and program operation was performed with fresh cell.

The model validation is performed using Synopsys TCAD simulations calibrated against previous retention modeling studies [4], [9]. In the model calculation, the trap energy was divided into 100 bins, and the CTN layer was discretized into 30 slices. The variation of the  $t_{TOX}$  showed the most dominance in detrapping time transient curve. This is because  $t_{TOX}$  directly controls the tunneling probability from CTN traps toward the channel side, making it the most sensitive geometrical parameter for the detrapping transient. In Fig. 3(b),  $t_{TOX}$  from 3 nm to 5 nm for both TCAD simulation and the model were performed, showing good match with each other. The model accurately reproduces the oxide-thickness-dependent retention transient observed in the TCAD benchmark.

To verify the time-step-free formulation, an iterative refresh test was performed as shown in Fig. 3(c). In the refresh calculation, the trapped-charge distribution is sequentially updated after each retention interval and used as the initial condition for the next step, whereas the proposed method directly predicts the transient from the initial distribution without intermediate updates. Both approaches show nearly identical  $\Delta V_{th}$  behavior over the entire retention range, with only about 2% difference at  $10^9$  s.

Table I shows the comparison between TCAD and the model. The table shows how much computation time and resource efficiency could be improved through the time-step-free model. Since the model's computation time is independent from sequential time step, can be adopted to both short- and long-term retention operation based on given physical time.

### IV. CONCLUSION

This study presents a time-step-free physics-based modeling for retention loss due to detrapping of trapped electrons. The novelty of time-step-free modeling is demonstrated through significant computational optimization, achieving runtime independent of final physical time. Additionally, compared to TCAD simulations calibrated with physical parameters from various prior studies including cryogenic condition, the proposed model demonstrates good match with TCAD simulation within extremely short timeframe.

## REFERENCES

- [1] A. Goda, K. K. Muchherla, and P. Feeley, "Reliability of 3D NAND Flash for Future Storage Systems (Invited)," 2023 IEEE International Reliability Physics Symposium (IRPS), Monterey, CA, USA, 2023, pp. 1-10, doi: 10.1109/IRPS48203.2023.10118280.
- [2] D. G. Refaldi, G. Malavena, L. Chiavarone, N. Gagliuzzi, A. S. Spinelli, and C. M. Compagnoni, "Cryogenic Investigation of Vertical Charge Loss in 3D NAND Flash Memories," 2025 IEEE International Reliability Physics Symposium (IRPS), Monterey, CA, USA, 2025, pp. 1-7, doi: 10.1109/IRPS48204.2025.10983672.
- [3] D. G. Refaldi, G. Malavena, L. Chiavarone, A. S. Spinelli, and C. M. Compagnoni, "Evidence of Widely Distributed Time Constants in the Vertical Charge Loss of 3-D Charge-Trap NAND Flash Memories," in IEEE Electron Device Letters, vol. 45, no. 10, pp. 1811-1814, Oct. 2024, doi: 10.1109/LED.2024.3435345.
- [4] M. Jin and H. Shin, "Modeling Attempt-to-Escape Frequency: Tunneling Emission of Trapped Electrons in Tunneling Oxides of 3-D NAND Flash Memory," in IEEE Transactions on Electron Devices, vol. 72, no. 9, pp. 4884-4889, Sept. 2025, doi: 10.1109/TED.2025.3589340.
- [5] S. M. Amoroso, C. Monzio Compagnoni, A. Mauri, A. Maconi, A. S. Spinelli, and A. L. Lacaita, "Semi-Analytical Model for the Transient Operation of Gate-All-Around Charge-Trap Memories," in IEEE Transactions on Electron Devices, vol. 58, no. 9, pp. 3116-3123, Sept. 2011, doi: 10.1109/TED.2011.2159010.
- [6] C. Lee and H. Shin, "Investigation and modeling of vertical redistribution in charge-trap-based 3D NAND Flash memory," Japanese Journal of Applied Physics 65.1 (2026): 011004, doi: 10.35848/1347-4065/ae3161
- [7] H. Choi, J. Yoo, and H. Shin, "A New Physical Model for Program Transients of Cylindrical Charge-Trap-Based NAND Flash Memories," in IEEE Transactions on Electron Devices, vol. 71, no. 4, pp. 2386-2392, April 2024, doi: 10.1109/TED.2024.3364587.
- [8] D. Verreck, A. Arreghini, F. Schanovsky, G. Rzepa, Z. Stanojevic, F. Mitterbauer, C. Kernstrook, O. Baumgartner, M. Karner, G. Van den Bosch, and M. Rosmeulen, "Understanding the ISPP Slope in Charge Trap Flash Memory and its Impact on 3-D NAND Scaling," 2021 IEEE International Electron Devices Meeting (IEDM), San Francisco, CA, USA, 2021, pp. 1-4, doi: 10.1109/IEDM19574.2021.9720506.
- [9] I. Han and H. Shin, "Machine Learning-Based Investigation of Temperature Effects on Lateral Migration in 3-D nand Flash Memory," in IEEE Transactions on Electron Devices, vol. 73, no. 5, pp. 2747-2753, May 2026, doi: 10.1109/TED.2026.3673641.
- [10] TCAD Sentaurus Device User Guide Version 2022, Synopsys, Mountain View, CA, USA, Mar. 2022.